**ZP-J Wheel Illustrated Companion**

# Dividing by Zero

*What a Wheel Is, and Why /0 Becomes a Real Element*

ZP Companion | The Wheel of Fractions | Initial: June 2026 | Current: 1.18, November 2026

This companion explains in plain language the ZP-J Wheel Addendum: the machine-verified proof that the "wheel of fractions" is a wheel - an algebraic structure (Carlström, 2001) in which dividing by zero is a defined operation rather than an error. Every formal claim here restates a theorem already proved in the technical document and its Lean source (ZeroParadox/Algebra/Wheel.lean, ZeroParadox/Algebra/WheelFrac.lean). Analogies are included to build intuition, not as proof claims. Consult the addendum for the authoritative mathematics.

## The Problem With Dividing by Zero

In ordinary arithmetic, 1/0 is undefined. You cannot simply declare it equal to some number: every attempt breaks one of the laws of arithmetic. If $1/0 = \infty$ and you insist on the usual rules, you quickly derive contradictions like $1 = 2$. So mathematics normally leaves division by zero out, and "/" stays a partial operation - defined for every input except one.

A wheel is a structure that makes division total instead. Every element, including 0, gets a reciprocal /x, so /0 becomes a first-class member of the system rather than an error. The trick is not to force /0 to be an ordinary number - it is to add new elements and adjust the laws so that nothing breaks. The name is Setzer's, and Carlström records where it came from: the term was "inspired by the topological picture ⊙ of the projective line together with an extra point 0/0" - a circle for the line, and the dot for the extra element.

## The Two New Elements: ∞ and ⊥

Making division total introduces exactly two new elements:

$\infty = /0$ (the reciprocal of zero). This is the "point at infinity" you may have met on the Riemann sphere, where 0 and ∞ sit at opposite poles. A wheel goes one step further than the sphere - it adds the second element ⊥ below as well - and it makes ∞ a genuine element you can compute with.

$\bot = 0 \cdot /0$ (zero times the reciprocal of zero). This is an absorbing "undefined" element: combine it with anything and you get ⊥ back. It is the wheel's honest answer to a genuinely meaningless expression - instead of crashing, the arithmetic returns a dedicated value that means "no information." (The wheel writes this element ⊥. Carlström (printed p. 3) traces the pair to Edalat and Potts, who suggested adjoining 1/0 and 0/0 to the reals, obtaining what domain theory calls the "lifting" of the real projective line: 1/0 is the projective line's point at infinity, and 0/0 is the bottom the lifting adds. The Zero Paradox uses the same symbol for the bottoms of its own carriers; the last section says what the wheel's element shares with them and what it does not.)

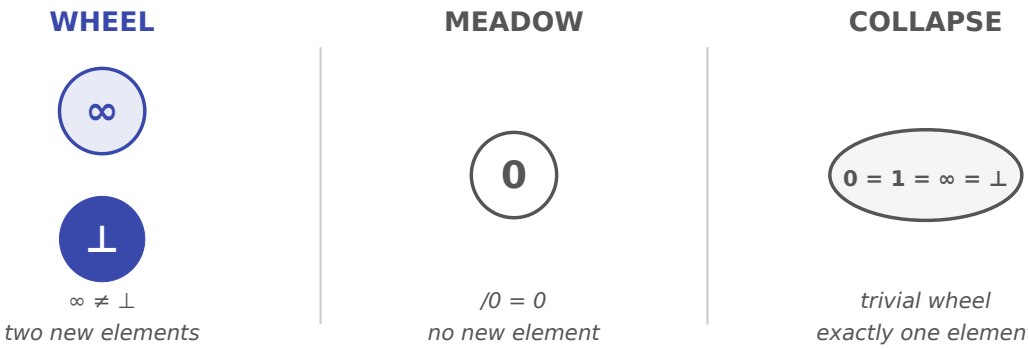
**Real-world analogy - The calculator that never crashes**

A pocket calculator shows "Error" when you divide by zero, and then you have to clear it before you can do anything else. A wheel is like a calculator that, instead of locking up, returns a special "undefined" reading - and lets you keep computing. Any further operation on that reading just returns "undefined" again. Nothing is lost and nothing breaks; the error is absorbed into a value the system knows how to carry.

## Wheel or Meadow? The Key Question

There are two established ways to make division by zero total, and they differ in what they add. A wheel adds the two new elements above and keeps them apart. A meadow adds nothing at all: it stays inside the number system you started with and simply declares $/0 = 0$. Bergstra, Hirshfeld and Tucker define a meadow as "a commutative ring with a total inverse operator satisfying two equations which imply $0^{-1} = 0$", so there is no second new element in a meadow for $\infty$ to be equal to. Its expressions still meet: $1/0$ and $0 \cdot (1/0)$ both come out as the ordinary 0, so a meadow identifies the two expressions a wheel keeps apart, which no wheel with more than one element can do (Carlström, Proposition 4.4).

That leaves the obvious follow-up: what if you took a wheel and identified its two new elements anyway? The answer is that you do not get a third structure - you get the trivial one. By Carlström's Proposition 4.4, if any two of 0, 1, /0 and $0 \cdot /0$ are equal in a wheel, then every element equals every other element and the wheel has exactly one element. So $\infty \neq \bot$ is not a preference; it is the price of having anything to compute with.




Three answers to 1/0, compared by the elements each one has. The wheel adjoins two and keeps them distinct. The meadow adjoins none, declaring zero its own reciprocal. Identifying the wheel's two elements is not a third algebra: it collapses everything into a single element (Carlström, Proposition 4.4).

## Building It: Fractions as Pairs

The construction at the heart of the addendum is the wheel of fractions, written $\odot_S A$. It generalises the way you build fractions from whole numbers, but it is designed to survive division by zero. Start with any commutative ring A (a number system with + and ×, like the integers) and a set S of "allowed denominators" closed under multiplication. Write each fraction as a pair (x, y), meaning x/y.

Reciprocal is just swapping the pair: $/(x, y) = (y, x)$. That single move is what makes division total - and it is why /0 exists. The fraction 0 is (0, 1), so /0 swaps to $(1, 0) = \infty$, and $0 \cdot /0$ works out to $(0, 0) = \bot$. Nothing is forbidden; the pair (1, 0) is a perfectly good fraction in this system.

There is one genuine subtlety: when are two fractions equal? The schoolbook rule - cross-multiply, $x \cdot y' = x' \cdot y$ - does not work on a general number system, because it fails to be transitive without a cancellation law (you cannot always divide out a common factor). The wheel of fractions fixes this by requiring a witness from S for every equality: two fractions count as equal only when elements of S certify it. Checking that this repaired notion of equality behaves - and that $+$, $\times$, and $/$ are well defined on it - is the bulk of the formal proof.

## The Main Result

The headline theorem is that this construction always works. For any commutative ring A and any choice of allowed denominators S, the wheel of fractions $\odot_S A$ satisfies every wheel axiom - all eight of Carlström's Definition 1.1 (checked in Lean as 14 separate equational fields). It is a wheel, with no exceptions and no extra hypotheses.

**ZeroParadox.instWheel (machine-verified, Lean 4)**

For every commutative ring A and every multiplicative submonoid S, the wheel of fractions $\odot_S A$ is a wheel - it satisfies all of Carlström's axioms. The proof is sorry-free and free of the axiom of choice: its only foundations are propositional extensionality and quotient soundness ([propext, Quot.sound]), both standard in Lean 4.

## And Not the Trivial Wheel

Being a wheel is only half the story. The other half is that this construction is a non-trivial wheel - the one where $\infty$ and $\bot$ stay distinct - and not the one-element collapse. The addendum proves one direction of that: as long as 0 is not one of the allowed denominators ($0 \notin S$), the reciprocal of zero and the absorbing element are different.

That condition is the normal situation: any sensible set of denominators excludes 0. The nonzero elements of an integral domain, or the complement of a prime ideal (the usual choice when building fractions), all avoid 0 - so the two elements stay apart. That deliberately allowing 0 as a denominator collapses the whole thing to a single element is Carlström's own observation (printed p. 5), not a result of this corpus.

**ZeroParadox.inf_ne_bot (machine-verified, Lean 4)**

If $0 \notin S$, then $\infty \neq \bot$ in $\odot_S A$. The construction is therefore a non-trivial wheel. (If $\infty$ and $\bot$ were equal, the witnessing elements of S would force 0 itself into S, contradicting the hypothesis.) Also sorry-free and choice-free: [propext, Quot.sound].

## Why This Belongs in the Zero Paradox

The Zero Paradox studies one role, written in several mathematical languages at once, each with its own occupant and no equation across them: in the 2-adic valuation the occupant is 0, where $v_2(0) = \infty$; in non-well-founded set theory it is the Quine atom (CC-2), a set $x = \{x\}$ (Aczel 1988, Example 1.3); in a lattice under a self-application map it is $\bot$ of the lattice. The wheel of fractions is the algebraic face of that role. Where ZP-J says "the bottom is where measure runs to infinity," the wheel adjoins two elements at division by zero: $\infty = /0$, which inversion exchanges with 0 ($/\infty = 0$), and the absorbing $\bot = 0 \cdot /0$, distinct from $\infty$ in

the standard case where $0 \notin S$. Of the two, $\bot = 0{\cdot}/0$ is the one that steps only to itself under inversion; $\infty$ steps to 0 (see the last section). That exchange of 0 and $\infty$ under inversion is winv_wheelInf ($/\infty = 0$), with $\infty = /0$ by definition.

One honest limitation. The wheel is built on top of a ring you supply - the ring structure is an input, not something derived from the Zero Paradox's own axioms. The valuation condition $\mathrm{val}(0) = \top$ (the wvs_val_zero field of WheelValuationStructure) is assumed, motivated and not forced, as the Lean source says (ZeroParadox/Algebra/Wheel.lean §VIII). Given that the valuation turns products into sums, the condition holds exactly when the valuation is not zero everywhere (the examples closing §VII-b there). The wheel shows the algebraic face exists and is consistent; it does not claim to force it into being.

## What This Is, and What It Is Not

The wheel of fractions and the theorem that it is a wheel are Carlström's, not new mathematics. What the Zero Paradox adds is a faithful, machine-checked encoding - readable field by field against Carlström's Definition 1.1, with a verified axiom footprint showing it needs no axiom of choice - placed in the context of the Zero Paradox. The takeaway: division by zero is not an error to be avoided but a defined element to be understood, $\infty = /0$, which inversion exchanges with 0 (winv_wheelInf). The absorbing element $\bot = 0{\cdot}/0$ steps only to itself under inversion (wheelBot_selfOnlyAt_winv) and absorbs every point under addition (wheelBot_absorbs), so it is one occupant of the Quine-atom role in the wheel. Under addition it is the only one, since every point steps to it (wheelBot_addSelfOnly_unique); under inversion it is not the only one: 1 also steps only to itself under inversion (winv_one, the inverse of 1 is 1), and in any wheel with more than one element 1 and $0{\cdot}/0$ differ (Carlström, Proposition 4.4). Occupants in the framework's other carriers fill the same role and are not one object with it, since an equation across carriers is ill-typed. Reading the role as filled per carrier, rather than by one object, applies the structuralist reading of Benacerraf, "What Numbers Could Not Be," Philosophical Review 74(1) (1965) 47–73, p. 70 (his subject is numbers) to the Quine-atom role; the per-carrier results are an instance of that reading, not its source.